



Heart rate variability and cognitive processing: The autonomic response to task demands



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ABSTRACT

This study investigated variations in heart rate variability (HRV) as a function of cognitive demands. Participants completed an execution condition including the psychomotor vigilance task, a working memory task and a duration discrimination task. The control condition consisted of oddball versions (participants had to detect the rare event) of the tasks from the execution condition, designed to control for the effect of the task parameters (stimulus duration and stimulus rate) on HRV. The NASA-TLX questionnaire was used as a subjective measure of cognitive workload across tasks and conditions. Three major findings emerged from this study. First, HRV varied as a function of task demands (with the lowest values in the working memory task). Second, and crucially, we found similar HRV values when comparing each of the tasks with its oddball control equivalent, and a significant decrement in HRV as a function of time-on-task. Finally, the NASA-TLX results showed larger cognitive workload in the execution condition than in the oddball control condition, and scores variations as a function of task. Taken together, our results suggest that HRV is highly sensitive to overall demands of sustained attention over and above the influence of other cognitive processes suggested by previous literature. In addition, our study highlights a potential dissociation between objective and subjective measures of mental workload, with important implications in applied settings.

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1. Introduction

A large body of research has shown a direct link between cognitive processing and the cardiovascular system through autonomic vagal control (Thayer & Lane, 2009). A simple way of measuring that relationship is to look at heart rate variability (HRV), a non-invasive measurement of the interactions between the autonomic nervous system and the cardiovascular system, based on the study of oscillations of the interval between heartbeats (Malik et al., 1996; Pumpura, Howorka, Groves, Chester, & Nolan, 2002).

Thayer et al. have recently proposed the Neurovisceral Integration Model to account for the link between cognitive processing and the functioning of the autonomous nervous system (Thayer, Hansen, Saus-Rose, & Johnsen, 2009; Thayer & Lane, 2009). They pointed out that HRV is a particularly sensitive index of the changes in a flexible neural network that is dynamically organized in

response to situational requirements. The authors highlighted the role of the prefrontal cortex in the modulation of subcortical cardio-acceleratory circuits via an inhibitory pathway that is associated with vagal function and that can be indexed by HRV. The link of the frontal cortex to autonomic motor circuits responsible for both the sympathoexcitatory and parasympathoinhibitory effects on the heart seems to be controlled both by direct and indirect pathways. In this sense, one of the potential mediators underlying variations in HRV as a function of cognitive demands is the baroreceptor system, i.e., the negative feedback loop adjusting heart activity to blood pressure fluctuations. In fact, the baroreflex function appears to be influenced by specific behavioral manipulations of cognitive demands and mental workload (e.g., Duschek, Werner, & Reyes del Paso, 2013; Paso, González, & Hernández, 2004). Consequently, variations in the baroreflex function may therefore also mediate modulations in HRV during the specific task conditions. In any case, HRV is thought of as an overall index of central-peripheral neural feedback and central nervous system–autonomic nervous system integration (Thayer & Lane, 2000, 2009).

A cursory look to the literature on the relationship between HRV cognition shows that researchers have used a wide range of tasks,

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tapping different cognitive processes, which make it difficult to establish a finer-grained relationship between HRV and cognitive processing. In more specific terms, a number of studies have singled out a subset of mental workload components – executive demands – as key to understand the HRV-cognitive processing link, with lower HRV as executive demands increase (Backs & Seljos, 1994; Duschek, Muckenthaler, Werner, & Reyes del Paso, 2009; Hansen, Johnsen, & Thayer, 2003; Luft, Takase, & Darby, 2009; Mathewson et al., 2010; Mulder & Mulder, 1981). In this scenario, the above mentioned Neurovisceral Integration Model predicts an inverse relationship between executive task demands and levels of HRV, which seems to be confirmed by the studies cited above. However, the results of other studies appear to challenge this straightforward view of the relationship between HRV and cognitive processing. For instance, Fairclough and Houston (2004) failed to show differences between the congruent and incongruent conditions when participants had to name the color of the ink in a Stroop task, a well-known executive task (e.g., Egner & Hirsch, 2005). On the contrary, they showed that HRV was sensitive to time-on-task, pointing to the role of overall attention demands on HRV. In this same line, Chang and Huang (2012) showed that HRV varied as a function of attentional demands in a visual search task, with lower HRV in a conjunction search task than in a feature search task and a control condition in which participants passively watched to the stimuli.

Together with attention demands, perceptual difficulty seems to be another key factor modulating HRV. The results of two recent studies point in that direction. Chen, Tsai, Biltz, Stoffregen, and Wade (2015) reported lower HRV as a function of perceptual difficulty, but not as a function of working memory load (linked to executive functioning, e.g., Duncan & Owen, 2000). Particularly relevant here is the study by Luque-Casado, Zabala, Morales, Mateo-March, and Sanabria (2013), who compared HRV during performance of three tasks, tapping three different cognitive functions: the psychomotor vigilance task (PVT; a vigilance task), an endogenous temporal orienting task (a cognitive control task), and a duration discrimination task (a perceptual task). The results showed lower levels of HRV in the perceptual task than in the other two tasks, with no significant differences in the main indexes of HRV between the PVT and the temporal orienting task. In addition, they showed that HRV decreased with time-on-task, a result that did not seem to depend on the particular task running at that moment.

Overall, the outcome of the above-mentioned studies seem to nuance Thayer et al.'s Neurovisceral Integration Model, and point to some aspects of cognitive demand (i.e., perceptual difficulty and sustained attention) and not others (working memory i.e., workload, interference) as key task features modulating HRV. However, as Luque-Casado et al. acknowledged in their article, brain structures typically associated with executive processing seem to be also involved in difficult perceptual discrimination (Duncan & Owen, 2000). Thus, the question remains of whether a task purposely developed to involve high executive demands would induce a larger reduction in HRV than the perceptual task used by Luque-Casado et al. (2013).

The present study is aimed at further investigating the role of particular processing demands involved in task effects on HRV. We partially replicated Luque-Casado et al.'s (2013) manipulation, using the PVT and the duration discrimination task, but replaced the temporal orienting task by a N-back task. The N-back task tackles working memory capacity, a core component of executive functioning, by asking participant to tag and update short-term stored information on a trial-by-trial basis (Kirchner, 1958; Owen, McMillan, Laird, & Bullmore, 2005). Importantly, along with these three tasks, we included three parallel oddball tasks, with the same stimuli parameters for each of the three, but in which participants just had to detect a rare event within a sequence of frequent stimuli.

The inclusion of the oddball condition allowed us to control for an important aspect that has been neglected in the majority of previous studies investigating the relationship between HRV and cognitive processing: the potential influence of stimulus parameters of the task on the relationship between autonomic response and cognitive performance. That is, whether stimulus setting features (e.g., stimulus duration, inter-stimulus interval) may explain (at least partially) the influence of task performance on autonomic reactivity over and above any specific cognitive process (e.g., executive processing, memory, etc.) specifically tapped by the task. In this sense, to the best of our knowledge, the only task feature that has been investigated in relation to this issue is the motor activity during the cognitive task (Bush, Alkon, Obradović, Stamperdahl, & Boyce, 2011; Porges et al., 2007). While Porges et al. showed that only gross motor activity (e.g., bike pedaling) could modulate the relationship between autonomic response and cognitive processing, Bush et al. found changes on autonomic reactivity to various cognitive tasks that were related to the particular motor activity during each procedure. Here, by asking participants to perform an oddball version of the three main cognitive tasks we controlled for variations in HRV due to the particular stimulus features of the tasks (e.g., stimulus duration) regardless of the task demands, whilst largely reducing the motor activity.

Here, as a cross-task and cross-condition manipulation check, subjective mental load was assessed with the National Aeronautics and Space Administration Task Load Index (NASA-TLX) questionnaire (Hart & Staveland, 1988). The NASA-TLX sensitivity to mental workload has been demonstrated to be useful in a variety of cognitively demanding tasks such as aircraft piloting (Karavidas et al., 2010; Ma et al., 2014), air traffic control (Brookings, Wilson, & Swain, 1996), surgery (Zheng et al., 2012), or laboratory tasks context (Muth, Moss, Rosopa, Salley, & Walker, 2012). With the inclusion of the NASA-TLX we aimed at comparing objective (HRV) and subjective potential indices of mental load induced by the different task demands. This is not trivial since previous research has questioned the validity of subjective measures of mental load (see Annett, 2002 for discussion on this issue).

On the basis of Luque-Casado et al.'s (2013) findings and the previous related research, we expected the N-back task to exert a stronger modulation over HRV than the PVT. The question of interest was to see whether the N-back task would also influence HRV to a greater extent than the duration discrimination task, a result that would add further support to the Neurovisceral Integration Model. Importantly, given that the three tasks in the oddball condition were essentially the same task (with variations only in stimuli parameters) with minimal response requirements, we did not expect significant differences in HRV across them. We predicted the NASA scores to parallel the HRV results, with larger perceived workload in the N-back task than in the other two tasks, and no differences across the three oddball tasks.

2. Methods and design

2.1. Participants

Twenty-four males undergraduate students (age range: 18–28 years old; $M = 21$ years old; $SD = 2.6$ years old) from the University of Granada (Spain) took part in the study in exchange of course credits. In order to take part in the experiment, participants were required to maintain a regular sleep–wake cycle for at least one day before the study and to abstain from stimulating beverages or any intense physical activity for the day of the experiment. Once in the laboratory, none of them reported having had any stimulating beverage or exercise session, and they all reported a regular sleep the night before (6–10 h; $M = 7.5$; $SD = 0.9$). None of the participants

smoked, and all of them reported normal hearing and normal or corrected-to-normal vision.

The experiment was approved by the local ethics committee and complied with the ethical standards laid down in the 1964 Declaration of Helsinki. Participants read and signed an informed consent statement before the beginning of the experimental session. They were also informed about their right to leave the experiment at any time. All data were analyzed and reported anonymously.

2.2. Apparatus and materials

Participants were fitted with a Polar H3 heart rate sensor and a Polar RS800CX monitor (Polar Electro Oy, Kempele, Finland) to record their HRV during the experimental session. We used a digital thermo-hygrometer Inovalley 207H01 (Inovalley, Saint-Ouen-l'Aumône, France) to measure the temperature and humidity percentage in the laboratory during the experimental session.

We used a PC Intel Quad Core i7-3770, a 24" LED monitor (BenQ XL2411T) and the E-Prime software (Psychology Software Tools, Pittsburgh, PA, USA) to control the stimulus presentation and response collection. The centre of the PC screen was situated at 60 cm (approx.) from the head of the participant and at his eye level. The PC keyboard was used to collect responses.

2.3. Procedure

The experimental protocol comprised two conditions (henceforth, execution and oddball). In the execution condition participants performed three different tasks: the PVT, a duration discrimination task and the N-back task. In the oddball condition, participants performed parallel 'oddball' versions of the three tasks mentioned above (see Section 2.3.1 for more details). The order of presentation of the execution and oddball conditions, and the tasks within each condition, were counterbalanced across participants. Immediately after each task, participants completed the NASA-TLX questionnaire (Hart & Staveland, 1988; Hart, 2006) for them to assess the subjective workload perceived for each task.

At the beginning of each experimental condition, all the participants had a familiarization period. They received verbal and written instructions and, after that, they practiced each task for one minute. They also received the necessary instructions to complete the NASA-TLX questionnaire at the beginning of the first experimental condition.

The timestamp of the start and end of each task was taken for further analysis of HRV. During the experiment, the participant was seated in front of the computer in a dimly illuminated room and isolated from external noise. Comfortable temperature ($21.3 \pm 0.8^\circ\text{C}$) and relative humidity ($43.4 \pm 3.1\%$) values were maintained throughout the experimental session.

2.3.1. Experimental tasks

2.3.1.1. Execution condition. PVT: We used a modified version of the task created by Wilkinson and Houghton (1982). On each trial, the number 3 in white color ($2.67^\circ \times 1.62^\circ$) appeared on the centre of the screen in a black background. Later, in a random time interval (from 2000 to 10,000 ms), this number changed its orientation from vertical to horizontal ($1.62^\circ \times 2.67^\circ$). The participants were instructed to respond with their dominant hand as fast as they could when the change in orientation occurred. Feedback of the response time was displayed on the screen on each trial during 300 ms. The next trial began after 1800 ms. Response anticipations were considered as errors. Participants were allowed 1500 ms to respond. If a response was not made during this time, the message "You did not answer" appeared on the screen. The task comprised a

single block of 12 min and the total number of trials was 111 ± 3.4 on average.

2.3.2. Duration discrimination task

The duration discrimination task was a psychophysical task in which participants had to make temporal judgments regarding which of two visual stimuli were presented for a longer period of time (Paul et al., 2011). The task started with the presentation of a fixation point at the centre of the screen for a random duration between 500 and 1000 ms. The fixation point was the "+" symbol ($0.38^\circ \times 0.38^\circ$) that remained on and steady for the whole trial. Then, two consecutive visual stimuli were presented (the sample and the comparison stimuli) with a random time interval of 500–1000 ms between them. The sample stimulus was a white number 3 and the comparison stimulus a red number 3 ($2.67^\circ \times 1.62^\circ$, both stimuli). The duration of the sample stimuli was 350 ms. Duration of the comparison stimulus was manipulated using the method of constant stimuli, lasting for either 160, 260, 300, 340, 380, 420, 460 or 560 ms. Participants had 3000 ms to respond before the start of the next trial. Once the participant responded, a random inter-trial time of 500–1000 ms of duration was presented.

Participants were instructed to discriminate whether the duration of the comparison stimulus was shorter or longer than the duration of the sample stimulus. If the duration of the comparison stimulus was longer than the duration of the sample stimulus, the participant should respond by pressing the up arrow. Otherwise, the participants should press the down arrow. The comparison stimuli of varying duration were randomly intermixed across trials. Each of the comparison stimuli was presented, on average, a 12.5% of the total number of trials in the task. There was not feedback after each trial. The overall duration of the task was 12 min and the total number of trials was, on average, 177 ± 9.3 . In this case, accuracy was stressed over response speed.

2.3.3. N-back task

One of four digits (1, 2, 3 or 4; $2.67^\circ \times 1.53^\circ$; $2.67^\circ \times 1.62^\circ$; $2.67^\circ \times 1.62^\circ$ and $2.67^\circ \times 1.81^\circ$, respectively) was presented for 500 ms, followed by a fixed delay of 2500 ms. Participants had to respond, at any time during the presentation of the stimulus or the delay period, whether the current stimulus displayed on the screen was the same as the stimulus presented two trials before. If the stimulus on the screen matched the stimulus presented two trials before, the participant had to press the letter "S" key on the PC Keyboard. Otherwise, the participants had to press the letter "N" key. A new stimulus was presented every 3000 ms (i.e., 500 ms of stimulus presentation and 2500 ms of fixed delay). The digit appearing on each trial was randomly selected, which means that, on average, the current digit was the same as the one presented two trials earlier in 25% of the trials. There was not feedback after each trial. The overall duration of the task was 12 min and the total number of trials was 217 ± 1.6 on average. Accuracy was stressed over response speed.

2.3.3.1. Oddball condition. In this condition, instructions were essentially the same for the three tasks: to detect the presence of an infrequent stimulus presented amongst a series of frequent stimuli. Crucially, there were three versions of the oddball task, corresponding to each of the task procedures described for the execution condition. Hence, each oddball version shared all task parameters (i.e., the rate of stimulus appearance, physical characteristics of the stimuli and duration of each interval) with its corresponding "execution" task, with the sole difference being the tasks demands. In effect, participants were instructed to respond only when the oddball stimuli (i.e., the stimulus in a green color) appeared on the screen (5% of the trials) in all three versions of

the oddball task. The oddball stimuli could be displayed instead of the number 3 in landscape orientation in the oddball version of the PVT, instead of the comparison stimuli in the oddball version of the duration discrimination task and instead of any of the numbers in the oddball version of the N-back task. The oddball stimuli were randomly intermixed across trials. Response anticipations or no response to oddball stimuli were considered as errors. Feedback was provided only after incorrect answers. In these cases, the word “incorrect” in a red color was displayed on the screen during 300 ms. The overall duration of each task was 12 min and the total number of trials, on average, was 94 ± 2.5 for the oddball version of the PVT, 112 ± 1.1 for the oddball version of the duration discrimination task and 216 ± 0 for the oddball version of the N-back task. Accuracy was stressed over response speed in all cases.

2.3.4. HRV measures

The elastic electrode transmitter belt (Polar H3 heart rate sensor) was placed on the chest of the participant at the level of the lower third of the sternum (just below the chest muscles) with conductive gel being applied as described by the manufacturer. This transmitter belt contains two electrodes to detect the voltage differential on the skin during every heart beat and sends the signal continuously and wirelessly using an electromagnetic field to the Polar RS800CX receiver unit. The data were collected with a sampling rate of 1000 Hz, providing a temporal resolution of 1 ms for each RR interval. This Polar equipment has been shown to be a valid and highly reliable way to measure short-term HRV at rest (Nunan et al., 2009; Radespiel-Tröger, Rauh, Mahlke, Gottschalk, & Mück-Weymann, 2003).

All data sets were transferred to a password-protected PC under ASCII format via Polar-specific software (Polar® ProTrainer 5 software version 5.35.161). Subsequently, each RR interval file was analysed by means of the Kubios HRV Analysis Software 2.0 (Tarvainen, Niskanen, Lipponen, Ranta-aho, & Karjalainen, 2009; The Biomedical Signal and Medical Imaging Analysis Group, Department of Applied Physics, University of Kuopio, Finland).

The recordings were preprocessed to exclude artifacts by eliminating RR intervals which differed more than 25% from the previous and the subsequent RR intervals (Malik, Cripps, Farrell, & Camm, 1989). Removed RR intervals were replaced by conventional spline interpolation so that the length of the data did not change (i.e., resulting in the same number of beats). We used the smoothness prior method with a Lambda value of 500 to remove disturbing low frequency baseline trend components (Tarvainen, Ranta-aho, & Karjalainen, 2002).

2.3.5. NASA-TLX questionnaire

The NASA-TLX provides an overall workload score (from 0 to 100 points) based on a weighted average of ratings on six dimensions: mental demands, physical demands, temporal demands, own performance, effort, and frustration.

Participants were instructed to rate each dimension on a visual analog scale (from 0 to 100 points). After that, participants were presented with 15 paired comparisons of each dimension and asked to choose which of them had a greater impact on their performance. A specific weight to each dimension from 0 to 5 was applied from these comparisons. The rating of each dimension was then multiplied by its respective weight and the total score (the sum of the scores of each dimension) was divided by 15 (the total number of paired comparisons) to obtain the final workload score.

3. Design and data reduction

3.1. Behavioral

We performed descriptive analysis on the behavioral data with the sole purpose of checking the proper performance during the

experimental session. With this aim, we obtained the overall mean reaction times (RTs), the Just noticeable difference (JND) and the overall accuracy percentage, for the PVT, duration discrimination task and N-back task, respectively. For oddball tasks, the percentage of accuracy of response to oddball stimuli was calculated. Only the experimental blocks were included in the analysis in all tasks.

For the PVT trials, with RTs below 100 ms (0.9%) and anticipations (i.e., responses prior to the target presentation; 1.7%) were discarded from the analysis (Basner & Dinges, 2011). JNDs were calculated for each participant by subtracting the stimulus onset asynchrony (SOA) at which the best fitting line crossed the 0.75 point from the SOA at which the same line crossed the 0.25 point and dividing by two. Thus, $JND = SOA(75\%) - SOA(25\%)/2$ and $SOA(75\%) = (z\text{-score}(75\%) - b/m)$ and $SOA(25\%) = (z\text{-score}(25\%) - b/m)$ with b = intercept and m = slope. The result of the equation is $JND = .675/m$ (Coren, Ward, & Enns, 1999).

3.2. HRV and NASA-TLX

The analysis of the HRV focused on both the time and frequency domains. The root-mean-square difference of successive normal R-R intervals (rMSSD) and the proportion of NN50 (i.e., the number of pairs of successive NNs that differ by more than 50 ms) were used as indexes of vagal control within the time domain (Allen, Chambers, & Towers, 2007). The high frequency (HF; 0.15–0.40 Hz) was used as the index of vagal tone in the frequency domain (Berntson et al., 1997; Paso, Langewitz, Mulder, van Roon, & Duschek, 2013). The denotations and definitions for the HRV parameters in this paper follow the guidelines given in *Task force of the European society of cardiology* and the *North American society of pacing and electrophysiology* (Malik et al., 1996).

In order to investigate the time course of HRV during task performance, 3 intervals of 4' each were considered for the analysis of the HRV. The ln-transformed HRV data were analyzed through repeated measures analysis of variance (ANOVA) with the within-participants factors of condition (execution and oddball), task (PVT, duration discrimination task and N-back task) and time-on-task (1, 2, 3). The NASA-TLX data were analysed through repeated measures ANOVA with the within-participants factors of condition (execution and oddball) and task (PVT, duration discrimination task and N-back task). The effect sizes were reported by partial eta-squared (η_{partial}^2). Sphericity was tested by means of the Mauchly sphericity test and the Green-House Geisser correction was applied when violation of this assumption occurred (corrected values are reported).

4. Results

4.1. Behavioral results

The descriptive analyses conducted on behavioral data showed a normal and typical execution for all tasks. On the one hand, the overall mean RT for the PVT was 210.27 ± 26.89 ms, the JND for the duration discrimination task was 77.19 ± 10.90 ms and the overall accuracy percentage for the N-back task was 91.27 ± 6.07 . On the other hand, the percentage of accuracy of response to oddball stimuli was 98.61 ± 6.80 and 98.44 ± 7.65 for the version oddball of the PVT and N-back task, respectively. The percentage of accuracy for the oddball version of the duration discrimination task could not be calculated due to faulty response recording. However, there was no problem in carrying out the task by the participants.

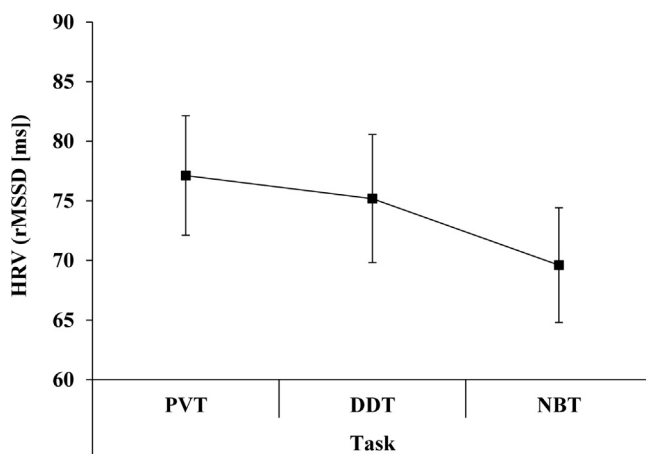
4.2. HRV

The ANOVAs showed that the main effect of condition (execution vs oddball) was not significant for any of the parameters, all

Table 1Mean ($\pm$ standard deviation) for the HRV parameters as a function of condition, task and time-on-task.

HRV parameters									
Task	rMSSD (ms)			pNN50 (%)			HF (ms ²)		
	Block 1	Block 2	Block 3	Block 1	Block 2	Block 3	Block 1	Block 2	Block 3
Execution condition									
PVT	82.4 (28.3)	76.9 (27.1)	79.2 (29.7)	46.2 (15.2)	43.9 (16.6)	43.0 (17.6)	2218.9 (1380.3)	1913.8 (967.0)	2130.6 (1536.8)
DDT	79.6 (32.2)	73.3 (31.0)	70.7 (27.0)	43.5 (18.7)	39.7 (18.1)	39.4 (15.4)	2762.4 (2651.3)	2135.8 (1646.0)	2062.9 (1568.9)
NBT	69.4 (29.0)	66.5 (26.3)	67.1 (23.2)	38.8 (18.5)	36.7 (17.5)	37.2 (17.0)	1965.1 (1661.1)	1626.2 (1170.4)	1749.0 (1221.5)
Oddball condition									
PVT	77.2 (26.4)	73.1 (24.3)	74.0 (24.5)	41.7 (15.3)	40.4 (15.7)	41.1 (14.2)	2175.8 (1497.6)	1893.2 (1305.1)	1935.1 (1102.5)
DDT	78.1 (33.1)	76.2 (33.5)	73.2 (28.9)	41.0 (16.1)	40.7 (16.7)	38.5 (15.3)	2565.9 (2558.3)	2387.4 (2093.2)	2086.6 (1843.2)
NBT	73.1 (27.9)	69.2 (25.7)	72.5 (27.3)	40.0 (16.6)	38.6 (14.7)	39.5 (15.6)	2034.9 (1635.2)	1797.7 (1678.2)	1900.5 (1645.1)

PVT: psychomotor vigilance task; DDT: duration discrimination task; NBT: n-back task; O-PVT, O-DDT and O-NBT: the oddball version of each task, respectively.

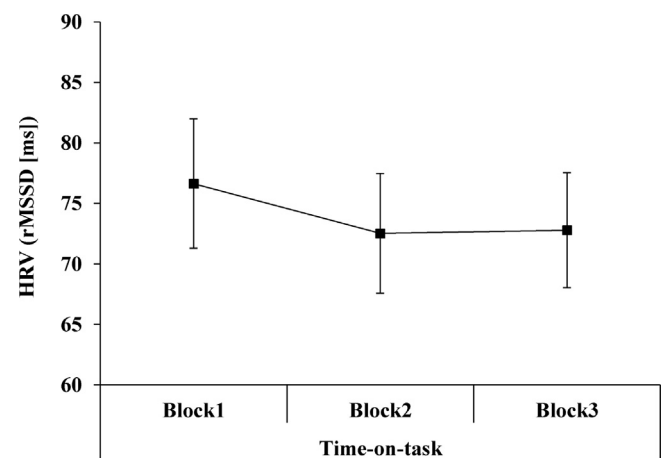
**Fig. 1.** HRV (rMSSD index) as a function of the task. The root-mean-square difference of successive normal R–R intervals (rMSSD) in milliseconds (ms) for each of the cognitive tasks (PVT = psychomotor vigilance task; DDT = duration discrimination task; NBT = n-back task). Bars represent standard errors of the mean.

$F_s < 1$ (see Table 1). On the contrary, the main effect of task reached statistical significance in all indexes, rMSSD, $F(2,46) = 10.52$, $p < .001$, $\eta_{\text{partial}}^2 = .31$ (see Fig. 1), pNN50, $F(2,46) = 5.98$, $p < .01$, $\eta_{\text{partial}}^2 = .20$, HF, $F(2,46) = 9.21$, $p < .001$, $\eta_{\text{partial}}^2 = .28$. All indexes showed the lowest values in the N-back task, all $p_s \leq .01$. The difference between the PVT and the duration discrimination task did not reach statistical significance in any of the parameters (all $p_s > .05$). HRV values in the duration discrimination task and the N-back tasks were significantly different for the rMSSD, $p < .01$, and HF, $p < .01$, but not for pNN50, $p = .053$. The main effect of time-on-task was significant for rMSSD, $F(2,46) = 6.51$, $p < .01$, $\eta_{\text{partial}}^2 = .22$ (see Fig. 2), and HF, $F(2,46) = 6.12$, $p < .01$, $\eta_{\text{partial}}^2 = .21$, and did not reach statistical significance for pNN50, $F(2,46) = 2.42$, $p = .10$. There were differences between block 1 and block 2 (rMSSD, $p < .001$, and HF, $p < .001$) and between block 1 and block 3 (rMSSD, $p = .01$ and HF, $p = .04$). The difference between block 2 and block 3 was not significant for any of the indexes, all $p_s > .42$.

Crucially, the interaction between task and condition did not reach statistical significance for any of the parameters, $F(2,46) = 2.71$, $p = .08$, $F(2,46) = 1.66$, $p = .20$, $F < 1$, for rMSSD, pNN50 and HF, respectively. The interactions between task and time-on-task, and between task, time-on-task and condition were not significant for any of the HRV indexes [all $F_s < 1$, except for task $\times$ time-on-task for HF, $F(2,46) = 1.63$, $p = .18$].

4.3. NASA-TLX scores

The repeated-measures ANOVA with the within-participants factors of condition (execution and oddball) and task (PVT, dura-

**Fig. 2.** HRV (rMSSD index) as a function of the time-on-task. The root-mean-square difference of successive normal R–R intervals (rMSSD) in milliseconds (ms) for each of the blocks of the three cognitive tasks (Block 1 = between 0 and 4 min of each task; Block 2 = between 4 and 8 min of each task; Block 3 = between 8 and 12 min of each task). Bars represent standard errors of the mean.**Table 2**Mean ($\pm$ standard deviation) for the NASA-TLX overall scores as a function of condition and task.

NASA-TLX overall scores		
Execution condition		
PVT	DDT	NBT
51.0 (23.8)	60.5 (16.0)	68.1 (14.4)
Oddball condition		
O-PVT	O-DDT	O-NBT
28.2 (18.6)	31.0 (21.8)	29.6 (20.3)

PVT: psychomotor vigilance task; DDT: duration discrimination task; NBT: n-back task; O-PVT, O-DDT and O-NBT: the oddball version of each task, respectively.

tion discrimination task and N-back task) revealed a significant main effect of condition, $F(1,23) = 76.36$, $p < .01$, $\eta_{\text{partial}}^2 = .77$, with greater scores in the execution condition than in the oddball condition (see Table 2). The main effect of task was also significant, $F(2,46) = 9.91$, $p < .01$, $\eta_{\text{partial}}^2 = .30$. Importantly, these main effects were better qualified by the significant interaction between condition and task, $F(2,46) = 6.29$, $p < .01$, $\eta_{\text{partial}}^2 = .21$ (see Fig. 3). In the execution condition, there were significant differences between the PVT and N-back task, $p < .01$, between the PVT and duration discrimination task, $p = .04$, and between the N-back and the duration discrimination task, $p = .03$. In all cases, the PVT and N-back task elicited the lowest and the highest scores respectively. Instead,

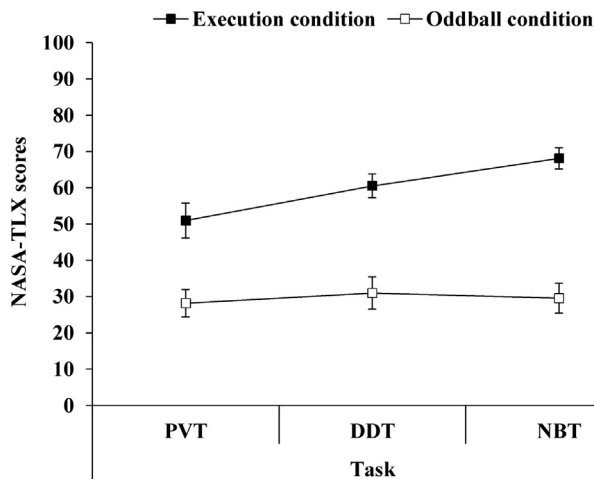


Fig. 3. NASA-TLX scores as a function of task for the execution and oddball condition. Mean of the NASA-TLX overall scores for the execution and oddball condition in each of the cognitive tasks (PVT = psychomotor vigilance task; DDT = duration discrimination task; NBT = n-back task). Bars represent standard errors of the mean.

comparisons between tasks for the oddball condition did not reveal significant differences (all $ps > .21$).

5. Discussion

HRV is sensitive to cognitive processing with significant variations as a function of changes in task demands. Previous accounts have pointed to executive demands of the task at the key parameter to explain those variations in HRV in the variety of tasks that have been tested in the laboratory (Thayer & Lane, 2009). On the contrary, other recent studies suggest that rather than broadly ranging executive demand, other, more molecular factors such as perceptual difficulty influence HRV. Here, we investigated this issue by comparing HRV values during performance of a vigilance task, a working memory task, and a duration discrimination task. Crucially, we also added a condition to control for the effect of the particular stimulus parameters of each task, minimizing the motor activity by using an oddball procedure. The NASA-TLX was used as a subjective measure of cognitive workload.

HRV indeed varied as a function of task demands, with lower values in the N-Back task than in the other two tasks, and no differences between the PVT and the duration discrimination task. At this point, these results would confirm the hypothesis stated by Luque-Casado et al. (2013) in their conclusions, whereby a task with greater executive demands than the temporal orienting task used in their study would induce lower HRV than the perceptual task.

However, the oddball control condition revealed intriguing results. The interaction between condition and task was not significant, apparently meaning that the reliable task effect was not (only) due to the particular demands of the tasks, since participants performed exactly the same task in the three version of the oddball procedure. It would however suggest that the particular stimulus features of the N-Back task, in comparison to the PVT and the duration discrimination task, were responsible for HRV decrements. The three tasks (in both the execution and oddball control conditions) were only differentiated in terms of stimuli duration, interval between stimuli, and, above all, presentation rate. The N-Back task, both in the execution and the oddball procedure, had twice the number of trials (in 12') as compared to the duration discrimination task and even more compared to the PVT, while the difference in the number of trials was much less between the PVT and the duration discrimination task (with no significant differ-

ences in HRV either). This larger number of trials in the N-Back task resulted in a larger number of motor responses, even in the oddball task. However, based on the scarce previous research (Bush et al., 2011; Porges et al., 2007) and given that the number of targets was very low in the oddball condition, one would argue that the motor demands of the task cannot explain the reliable task effect shown here. Note that if motor activity were responsible of the changes in HRV, significant differences would have emerged between the execution condition and the oddball condition for every task. On the contrary, it is more plausible that the higher presentation rate resulted, both in the execution and oddball conditions, in a significant increase in the demands of sustained attention (Lanzetta, Dember, Warm, & Berch, 1987; Parasuraman & Giambra, 1991; Sarter, Givens, & Bruno, 2001) with respect to the other two tasks, which in turn resulted in lower HRV. Crucially, two results from our study appear to support the hypothesis of the HRV sensitivity to sustained attention demands: the non-significant main effect of condition and the significant main effect of time-on-task.

The non-significant main effect of condition was driven by similar HRV values when comparing each of the tasks in the execution condition with their oddball equivalent. An oddball task like the one used here requires participants to maintain a high level of vigilance in order to detect the infrequent targets (Eason & Dudley, 1970; Fruhstorfer & Bergström, 1969) and it has been used as a paradigm to assess sustained attention (Czisch et al., 2012; Weber, Van Der Molen, & Molenaar, 1994). In addition, the robust time-on-task effect shown in our experiment is consistent with previous accounts suggesting the sensitivity of HRV to vigilance decrements or mental fatigue (e.g., Middleton, Sharma, Agouzoul, Sahakian, & Robbins, 1999; Porges & Raskin, 1969). For example, Luque-Casado et al. (2013), using a similar procedure to that of our execution condition, showed a gradual decrement in participants' HRV as a function of the time on task. Fairclough and Houston (2004) also showed that the 0.1 Hz component of HRV was sensitive to time on task although it did not seem sensitive to a manipulation of workload within the same task. Additionally, Chua et al. (2012) have recently shown that HRV provides information about a person's vigilance state, and that this measure could potentially be used to predict when an individual is at increased risk of attentional failure.

Taken together, our results therefore point to sustained attention demands of the tasks in general and of the NBack task in particular, as the major factor influencing HRV in our study, over and above any other of task-related cognitive components (e.g., working memory, cognitive control, perceptual processing) or task parameters.

Sustained attention has been considered one of the executive functions linked to the prefrontal cortex (Alvarez & Emory, 2006; Stuss, Shallice, Alexander, & Picton, 1995). In that sense, our results do not plainly contradict, but nuance the Neurovisceral Integration Model. Moreover, it would explain the outcome of previous research that failed to show differences between experimental conditions varying in terms of cognitive control demands (e.g., Fairclough & Houston, 2004; Hansen et al., 2003; Luque-Casado et al., 2013), and those reports showing that overall attentional load is crucial in order to explain HRV variations (e.g., Chang & Huang, 2012). Importantly, on the basis of our results, it would be interesting for future research to incorporate experimental designs including specific manipulations of the sustained attention load (e.g., by manipulating the likelihood of the target appearance) for a more thorough investigation of the specific effect of the sustained attention demands on HRV. In fact, this could be considered as a potential limitation of our study since the level of sustained attention load was not systematically manipulated in addition to task-related cognitive components or task parameters.

The results of the NASA-TLX replicated previous studies (e.g., Karavidas et al., 2010; Ma et al., 2014; Muth et al., 2012), show-

ing larger cognitive workload in the execution condition than in the oddball condition, and scores variations as a function of task, with larger values for the N-Back task, followed by the duration discrimination task and the PVT. Interestingly enough, NASA scores appear clearly differentiated from HRV measures, suggesting a dissociation between objective and subjective cognitive workload, and contributing to the discussion regarding the validity of subjective measures of cognitive workload (Annett, 2002). The present results show that two conditions that contribute differently to subjective load (which would suggest that one of them is much less loading, and, supposedly, less interfering or potentially dangerous than the other one), can actually be equally loading in psychophysiological terms. In other words, some aspects of mental workload can remain hidden to subjective insight.

In sum, the outcome of the present study suggests that HRV is highly sensitive to sustained attention over and above the influence of other cognitive processes, a finding that needs to be considered by any research looking at the link between autonomic control and cognitive processing. In addition, our study highlights a potential dissociation between objective and subjective measures of mental workload, which has important implications in applied settings.

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References

- Allen, J. J. B., Chambers, A. S., & Towers, D. N. (2007). The many metrics of cardiac chronotropy: a pragmatic primer and a brief comparison of metrics. *Biological Psychology*, 74(2), 243–262. <http://dx.doi.org/10.1016/j.biopsycho.2006.08.005>
- Alvarez, J. A., & Emory, E. (2006). Executive function and the frontal lobes: a meta-analytic review. *Neuropsychology Review*, 16(1), 17–42. <http://dx.doi.org/10.1007/s11065-006-9002-x>
- Annett, J. (2002). Subjective rating scales: science or art? *Ergonomics*, 45(14), 966–987. <http://dx.doi.org/10.1080/00140130210166951>
- Backs, R. W., & Seljós, K. A. (1994). Metabolic and cardiorespiratory measures of mental effort: the effects of level of difficulty in a working memory task. *International Journal of Psychophysiology*, 16(1), 57–68. [http://dx.doi.org/10.1016/0167-8760\(94\)90042-6](http://dx.doi.org/10.1016/0167-8760(94)90042-6)
- Basner, M., & Dinges, D. F. (2011). Maximizing sensitivity of the psychomotor vigilance test (PVT) to sleep loss. *Sleep*, 34(5), 581–591.
- Berntson, G. G., Eckberg, J. T., Eckberg, D. L., Grossman, P., Kaufmann, P. G., Malik, M., et al. (1997). Heart rate variability: origins, methods, and interpretive caveats. *Psychophysiology*, 34(6), 623–648.
- Brookings, J. B., Wilson, G. F., & Swain, C. R. (1996). Psychophysiological responses to changes in workload during simulated air traffic control. *Biological Psychology*, 42(3), 361–377. [http://dx.doi.org/10.1016/0301-0511\(95\)05167-8](http://dx.doi.org/10.1016/0301-0511(95)05167-8)
- Bush, N. R., Alkon, A., Obradović, J., Stamperdahl, J., & Boyce, W. T. (2011). Differentiating challenge reactivity from psychomotor activity in studies of children's psychophysiology: considerations for theory and measurement. *Journal of Experimental Child Psychology*, 110(1), 62–79. <http://dx.doi.org/10.1016/j.jecp.2011.03.004>
- Chang, Y.-C., & Huang, S.-L. (2012). The influence of attention levels on psychophysiological responses. *International Journal of Psychophysiology*, 86(1), 39–47. <http://dx.doi.org/10.1016/j.ijpsycho.2012.09.001>
- Chen, F. C., Tsai, C. L., Biltz, G. R., Stoffregen, T. A., & Wade, M. G. (2015). Variations in cognitive demand affect heart rate in typically developing children and children at risk for developmental coordination disorder. *Research in Developmental Disabilities*, 38, 362–371. <http://dx.doi.org/10.1016/j.ridd.2014.12.002>
- Chua, E. C.-P., Tan, W.-Q., Yeo, S.-C., Lau, P., Lee, I., Mien, I. H., et al. (2012). Heart rate variability can be used to estimate sleepiness-related decrements in psychomotor vigilance during total sleep deprivation. *Sleep*, 35(3), 325–334. <http://dx.doi.org/10.5665/sleep.1688>
- Coren, S., Ward, L. M., & Enns, J. T. (1999). *Sensation and perception*. Fort Worth: Harcourt Brace College Publishers.
- Czisch, M., Wehrle, R., Harsay, H. A., Wetter, T. C., Holsboer, F., Samann, P. G., et al. (2012). On the need of objective vigilance monitoring: effects of sleep loss on target detection and task-negative activity using combined EEG/fMRI. *Frontiers in Neurology*, 3. <http://dx.doi.org/10.3389/fneur.2012.00067>
- Duncan, J., & Owen, A. M. (2000). Common regions of the human frontal lobe recruited by diverse cognitive demands. *Trends in Neurosciences*, 23(10), 475–483. [http://dx.doi.org/10.1016/S0166-2236\(00\)01633-7](http://dx.doi.org/10.1016/S0166-2236(00)01633-7)
- Duschek, S., Muckenthaler, M., Werner, N., & Reyes del Paso, G. A. (2009). Relationships between features of autonomic cardiovascular control and cognitive performance. *Biological Psychology*, 81(2), 110–117. <http://dx.doi.org/10.1016/j.biopsycho.2009.03.003>
- Duschek, S., Werner, N. S., & Reyes del Paso, G. A. (2013). The behavioral impact of baroreflex function: a review. *Psychophysiology*, 50(12), 1183–1193. <http://dx.doi.org/10.1111/psyp.12136>
- Eason, R. G., & Dudley, L. M. (1970). Physiological and behavioral indicants of activation. *Psychophysiology*, 7(2), 223–232. <http://dx.doi.org/10.1111/j.1469-8986.1970.tb02228.x>
- Egner, T., & Hirsch, J. (2005). The neural correlates and functional integration of cognitive control in a Stroop task. *NeuroImage*, 24(2), 539–547. <http://dx.doi.org/10.1016/j.neuroimage.2004.09.007>
- Fairclough, S. H., & Houston, K. (2004). A metabolic measure of mental effort. *Biological Psychology*, 66(2), 177–190. <http://dx.doi.org/10.1016/j.biopsycho.2003.10.001>
- Fruhstorfer, H., & Bergström, R. M. (1969). Human vigilance and auditory evoked responses. *Electroencephalography and Clinical Neurophysiology*, 27(4), 346–355. [http://dx.doi.org/10.1016/0013-4694\(69\)91443-6](http://dx.doi.org/10.1016/0013-4694(69)91443-6)
- Hansen, A. L., Johnsen, B. H., & Thayer, J. F. (2003). Vagal influence on working memory and attention. *International Journal of Psychophysiology*, 48(3), 263–274.
- Hart, S. G. (2006). Nasa-task load index (NASA-TLX); 20 years later. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 50(9), 904–908. <http://dx.doi.org/10.1177/154193120605000909>
- Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (Task Load Index): results of empirical and theoretical research. In P. Hancock, & N. Meshkati (Eds.), *Human mental workload*. Amsterdam, North Holland: North Holland Press.
- Karavidas, M. K., Lehrer, P. M., Lu, S.-E., Vaschillo, E., Vaschillo, B., & Cheng, A. (2010). The effects of workload on respiratory variables in simulated flight: a preliminary study. *Biological Psychology*, 84(1), 157–160. <http://dx.doi.org/10.1016/j.biopsycho.2009.12.009>
- Kirchner, W. K. (1958). Age differences in short-term retention of rapidly changing information. *Journal of Experimental Psychology*, 55(4), 352–358. <http://dx.doi.org/10.1037/h0043688>
- Lanzetta, T. M., Dember, W. N., Warm, J. S., & Berch, D. B. (1987). Effects of task type and stimulus heterogeneity on the event rate function in sustained attention. *Human Factors*, 29(6), 625–633.
- Luft, C. D. B., Takase, E., & Darby, D. (2009). Heart rate variability and cognitive function: effects of physical effort. *Biological Psychology*, 82(2), 186–191. <http://dx.doi.org/10.1016/j.biopsycho.2009.07.007>
- Luque-Casado, A., Zabala, M., Morales, E., Mateo-March, M., & Sanabria, D. (2013). Cognitive performance and heart rate variability: the influence of fitness level. *PLoS One*, 8(2), e56935. <http://dx.doi.org/10.1371/journal.pone.0056935>
- Ma, J., Ma, R.-M., Liu, X.-W., Bian, K., Wen, Z.-H., Li, X.-J., et al. (2014). Workload influence on fatigue related psychological and physiological performance changes of aviators. *PLoS One*, 9(2), e87121. <http://dx.doi.org/10.1371/journal.pone.0087121>
- Malik, M., Bigger, J. T., Camm, A. J., Kleiger, R. E., Malliani, A., Moss, A. J., et al. (1996). Heart rate variability standards of measurement, physiological interpretation, and clinical use. *European Heart Journal*, 17(3), 354–381.
- Malik, M., Cripps, T., Farrell, T., & Camm, A. J. (1989). Prognostic value of heart rate variability after myocardial infarction. A comparison of different data-processing methods. *Medical and Biological Engineering and Computing*, 27(6), 603–611. <http://dx.doi.org/10.1007/bf02441642>
- Mathewson, K. J., Jetha, M. K., Drmic, I. E., Bryson, S. E., Goldberg, J. O., Hall, G. B., et al. (2010). Autonomic predictors of Stroop performance in young and middle-aged adults. *International Journal of Psychophysiology*, 76(3), 123–129. <http://dx.doi.org/10.1016/j.ijpsycho.2010.02.007>
- Middleton, H. C., Sharma, A., Agouzoul, D., Sahakian, B. J., & Robbins, T. W. (1999). Contrasts between the cardiovascular concomitants of tests of planning and attention. *Psychophysiology*, 36(5), 610–618.
- Mulder, G., & Mulder, L. J. M. (1981). Information processing and cardiovascular control. *Psychophysiology*, 18(4), 392–402. <http://dx.doi.org/10.1111/j.1469-8986.1981.tb02470.x>
- Muth, E. R., Moss, J. D., Rosopa, P. J., Salley, J. N., & Walker, A. D. (2012). Respiratory sinus arrhythmia as a measure of cognitive workload. *International Journal of Psychophysiology*, 83(1), 96–101. <http://dx.doi.org/10.1016/j.ijpsycho.2011.10.011>
- Nunan, D., Donovan, G., Jakovljevic, D. G., Hodges, L. D., Sandercock, G. R. H., & Brodie, D. A. (2009). Validity and reliability of short-term heart-rate variability from the Polar S810. *Medicine and Science in Sports and Exercise*, 41(1), 243–250. <http://dx.doi.org/10.1249/mss.0b013e318184a4b1>
- Owen, A. M., McMillan, K. M., Laird, A. R., & Bullmore, E. (2005). N-back working memory paradigm: a meta-analysis of normative functional neuroimaging studies. *Human Brain Mapping*, 25(1), 46–59. <http://dx.doi.org/10.1002/hbm.20131>

- Parasuraman, R., & Giambra, L. (1991). *Skill development in vigilance: effects of event rate and age*. *Psychology and Aging*, 6(2), 155–169.
- Paul, I., Wearden, J., Bannier, D., Gontier, E., Le Dantec, C., & Rebaï, M. (2011). Making decisions about time: event-related potentials and judgements about the equality of durations. *Biological Psychology*, 88(1), 94–103. <http://dx.doi.org/10.1016/j.biopsycho.2011.06.013>
- Porges, S. W., Heilman, K. J., Bazhenova, O. V., Bal, E., Doussard-Roosevelt, J. A., & Koledin, M. (2007). Does motor activity during psychophysiological paradigms confound the quantification and interpretation of heart rate and heart rate variability measures in young children? *Developmental Psychobiology*, 49(5), 485–494. <http://dx.doi.org/10.1002/dev.20228>
- Porges, S. W., & Raskin, D. C. (1969). *Respiratory and heart rate components of attention*. *Journal of Experimental Psychology*, 81(3), 497–503.
- Pumprla, J., Howorka, K., Groves, D., Chester, M., & Nolan, J. (2002). Functional assessment of heart rate variability: physiological basis and practical applications. *International Journal of Cardiology*, 84(1), 1–14. [http://dx.doi.org/10.1016/s0167-5273\(02\)00057-8](http://dx.doi.org/10.1016/s0167-5273(02)00057-8)
- Radespiel-Tröger, M., Rauh, R., Mahlke, C., Gottschalk, T., & Mück-Weymann, M. (2003). Agreement of two different methods for measurement of heart rate variability. *Clinical Autonomic Research*, 13(2), 99–102. <http://dx.doi.org/10.1007/s10286-003-0085-7>
- Reyes del Paso, G. A., González, I., & Hernández, J. A. (2004). Baroreceptor sensitivity and effectiveness varies differentially as a function of cognitive-attentional demands. *Biological Psychology*, 67(3), 385–395. <http://dx.doi.org/10.1016/j.biopsycho.2004.02.001>
- Reyes del Paso, G. A., Langewitz, W., Mulder, L. J. M., van Roon, A., & Duschek, S. (2013). The utility of low frequency heart rate variability as an index of sympathetic cardiac tone: a review with emphasis on a reanalysis of previous studies. *Psychophysiology*, 50(5), 477–487. <http://dx.doi.org/10.1111/psyp.12027>
- Sarter, M., Givens, B., & Bruno, J. P. (2001). The cognitive neuroscience of sustained attention: where top-down meets bottom-up. *Brain Research Reviews*, 35(2), 146–160. [http://dx.doi.org/10.1016/s0165-0173\(01\)00044-3](http://dx.doi.org/10.1016/s0165-0173(01)00044-3)
- Stuss, D. T., Shallice, T., Alexander, M. P., & Picton, T. W. (1995). A multidisciplinary approach to anterior attentional functions. *Annals of the New York Academy of Sciences*, 769, 191–212. <http://dx.doi.org/10.1111/j.1749-6632.1995.tb38140.x>
- Tarvainen, M. P., Niskanen, J.-P., Lipponen, J. A., Ranta-aho, P. O., & Karjalainen, P. A. (2009). Kubios HRV—a software for advanced heart rate variability analysis. In J. Sloten, P. Verdonck, M. Nyssen, J. Haueisen, & R. Magjarevic (Eds.), *4th European Conference of the International Federation for Medical and Biological Engineering* (Vol. 22) (pp. 1022–1025). Berlin, Heidelberg: Springer.
- Tarvainen, M. P., Ranta-aho, P. O., & Karjalainen, P. A. (2002). An advanced detrending method with application to HRV analysis. *Biomedical Engineering, IEEE Transactions*, 49(2), 172–175. <http://dx.doi.org/10.1109/10.979357>
- Thayer, J. F., Hansen, A. L., Saus-Rose, E., & Johnsen, B. H. (2009). Heart rate variability, prefrontal neural function, and cognitive performance: the neurovisceral integration perspective on self-regulation, adaptation, and health. *Annals of Behavioral Medicine*, 37(2), 141–153. <http://dx.doi.org/10.1007/s12160-009-9101-z>
- Thayer, J. F., & Lane, R. D. (2000). A model of neurovisceral integration in emotion regulation and dysregulation. *Journal of Affective Disorders*, 61(3), 201–216. [http://dx.doi.org/10.1016/s0165-0327\(00\)00338-4](http://dx.doi.org/10.1016/s0165-0327(00)00338-4)
- Thayer, J. F., & Lane, R. D. (2009). Claude Bernard and the heart–brain connection: further elaboration of a model of neurovisceral integration. *Neuroscience & Biobehavioral Reviews*, 33(2), 81–88. <http://dx.doi.org/10.1016/j.neubiorev.2008.08.004>
- Weber, E. J. M., Van Der Molen, M. W., & Molenaar, P. C. M. (1994). Heart rate and sustained attention during childhood: age changes in anticipatory heart rate, primary bradycardia, and respiratory sinus arrhythmia. *Psychophysiology*, 31(2), 164–174. <http://dx.doi.org/10.1111/j.1469-8986.1994.tb01036.x>
- Wilkinson, R. T., & Houghton, D. (1982). *Field test of arousal: a portable reaction timer with data storage*. *Human Factors*, 24(4), 487–493.
- Zheng, B., Jiang, X., Tien, G., Meneghetti, A., Panton, O. N. M., & Atkins, M. S. (2012). Workload assessment of surgeons: correlation between NASA TLX and blinks. *Surgical Endoscopy*, 26(10), 2746–2750. <http://dx.doi.org/10.1007/s00464-012-2268-6>